Homework 09

Question 1: Choosing an SVM kernel

```
library(e1071)
set.seed(432)

sah <- read.csv("~/Desktop/stat432-fall2026/week-09/data/saheart.csv",
                stringsAsFactors = FALSE)
sah$famhist <- ifelse(sah$famhist == "Present", 2, 1)
sah$chd <- factor(sah$chd, levels = c(0, 1))

preds <- c("sbp", "tobacco", "ldl", "adiposity", "famhist",
           "typea", "obesity", "alcohol", "age")
X <- as.matrix(sah[, preds])
y <- sah$chd
n <- nrow(X)
```

Candidate settings

Sixteen candidates in total (at most 20 allowed). Held fixed for every candidate: `scale = FALSE` (I standardized manually inside each fold), `coef0 = 1` and `gamma = 1/9` (one over the number of predictors) for the polynomial kernels, and e1071's default tolerance and no class weights.

```
cand <- rbind(
  data.frame(kernel = "linear", degree = NA, C = c(0.01, 0.1, 1, 10),
             gamma = NA),
  expand.grid(kernel = "polynomial", degree = 2:4, C = c(0.1, 1),
              gamma = 1/9, stringsAsFactors = FALSE),
  expand.grid(kernel = "radial", degree = NA, C = c(1, 10),
              gamma = c(0.01, 0.05, 0.2), stringsAsFactors = FALSE)
```

```
)
cand$id <- seq_len(nrow(cand))
nrow(cand)
```

```
[1] 16
```

Folds and cross-validation

```
folds <- sample(rep(1:5, length.out = n))
fit_predict <- function(k, tr, te) {
  mu <- colMeans(X[tr, ])
  s  <- apply(X[tr, ], 2, sd)
  Xtr <- scale(X[tr, ], center = mu, scale = s)
  Xte <- scale(X[te, ], center = mu, scale = s)
  args <- list(x = Xtr, y = y[tr], type = "C-classification",
               kernel = cand$kernel[k], cost = cand$C[k], scale = FALSE)
  if (cand$kernel[k] == "polynomial") {
    args$degree <- cand$degree[k]; args$gamma <- cand$gamma[k]
    args$coef0 <- 1
  }
  if (cand$kernel[k] == "radial") args$gamma <- cand$gamma[k]
  predict(do.call(svm, args), Xte)
}

cv_err <- sapply(seq_len(nrow(cand)), function(k) {
  fold_err <- sapply(1:5, function(f) {
    te <- which(folds == f); tr <- which(folds != f)
    mean(fit_predict(k, tr, te) != y[te])
  })
  mean(fold_err)
})
cand$cv_error <- cv_err
```

```
res <- cand[, c("id", "kernel", "degree", "C", "gamma", "cv_error")]
res$cv_error <- round(res$cv_error, 4)
knitr::kable(res, row.names = FALSE,
             caption = "Mean 5-fold CV misclassification rate by candidate")
```

Table 1: Mean 5-fold CV misclassification rate by candidate

id	kernel	degree	C	gamma	cv_error
1	linear	NA	0.01	NA	0.2770
2	linear	NA	0.10	NA	0.2791
3	linear	NA	1.00	NA	0.2813
4	linear	NA	10.00	NA	0.2813
5	polynomial	2	0.10	0.1111111	0.2706
6	polynomial	3	0.10	0.1111111	0.2749
7	polynomial	4	0.10	0.1111111	0.2814
8	polynomial	2	1.00	0.1111111	0.2793
9	polynomial	3	1.00	0.1111111	0.3183
10	polynomial	4	1.00	0.1111111	0.3399
11	radial	NA	1.00	0.0100000	0.2619
12	radial	NA	10.00	0.0100000	0.2684
13	radial	NA	1.00	0.0500000	0.2728
14	radial	NA	10.00	0.0500000	0.3009
15	radial	NA	1.00	0.2000000	0.2879
16	radial	NA	10.00	0.2000000	0.3442

```
best <- which.min(cand$cv_error)
plot(cand$id, cand$cv_error, pch = 19, xlab = "Candidate id",
     ylab = "Mean CV misclassification rate",
     col = c(linear = "black", polynomial = "blue",
             radial = "red")[cand$kernel])
points(best, cand$cv_error[best], cex = 2)
legend("topleft", c("linear", "polynomial", "radial"),
       col = c("black", "blue", "red"), pch = 19)
```

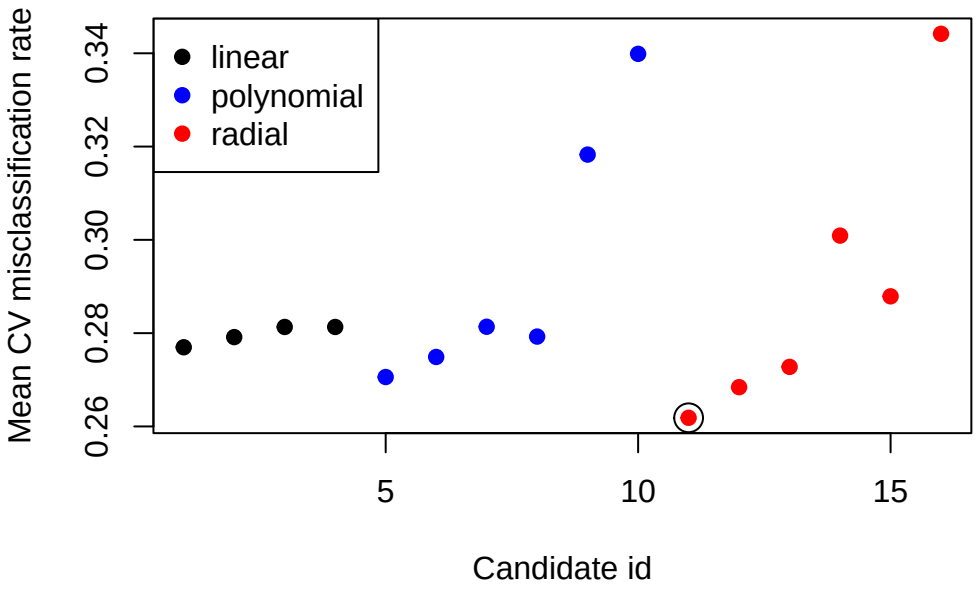

Figure 1: Mean 5-fold CV misclassification rate for the 16 candidates (circled: minimum, radial kernel with C = 1, gamma = 0.01).

Preferred model

```
b <- cand[best, ]
b
```

```
   kernel degree C gamma id  cv_error
11 radial     NA 1  0.01 11 0.2618513
```

```
best_by_kernel <- tapply(cand$cv_error, cand$kernel, min)
round(best_by_kernel, 4)
```

```
    linear polynomial     radial
    0.2770     0.2706     0.2619
```

I compared 16 SVMs (linear, polynomial of degree 2–4, and radial) over a grid of C and, for the radial kernel, γ, using the same five folds and standardizing with only the fitting observations in each fold (fixed: coef0 $= 1$, $\gamma = 1/9$ for polynomial kernels). The best mean CV misclassification rates were 0.277 (linear, $C = 0.01$), 0.271 (polynomial), and 0.262 (radial, $C = 1, \gamma = 0.01$). The differences between kernels are smaller than one standard error (≈ 0.02), so I prefer the simpler linear kernel with $C = 0.01$ (CV error 0.277), which is within about one standard error of the minimum ($0.262 + 0.02 \approx 0.282$); the strict minimizer is the radial kernel.

Question 2: Write and use a short SVM analysis skill

The one-page report produced with my skill, followed by the complete `SKILL.md`, is included on the next pages.

SVM classification of coronary heart disease (chd = 1), SAheart data

n = 462, p = 9; stratified 70/30 split (train 323, test 139); 5−fold CV; seed 432.

Methods

Predictors were standardized (mean 0, SD 1) within each training fit; famhist was coded 1 = Absent, 2 = Present. Linear and RBF kernels were compared because a linear boundary is the simplest option and the RBF allows a curved one. The grid was C in {0.01, 0.1, 1, 10} and gamma in {0.01, 0.1, 1} (RBF only), 16 candidates, evaluated with one fixed stratified 5−fold CV. The lowest mean CV error was selected and the model was scored once on the test split.

Tuning results

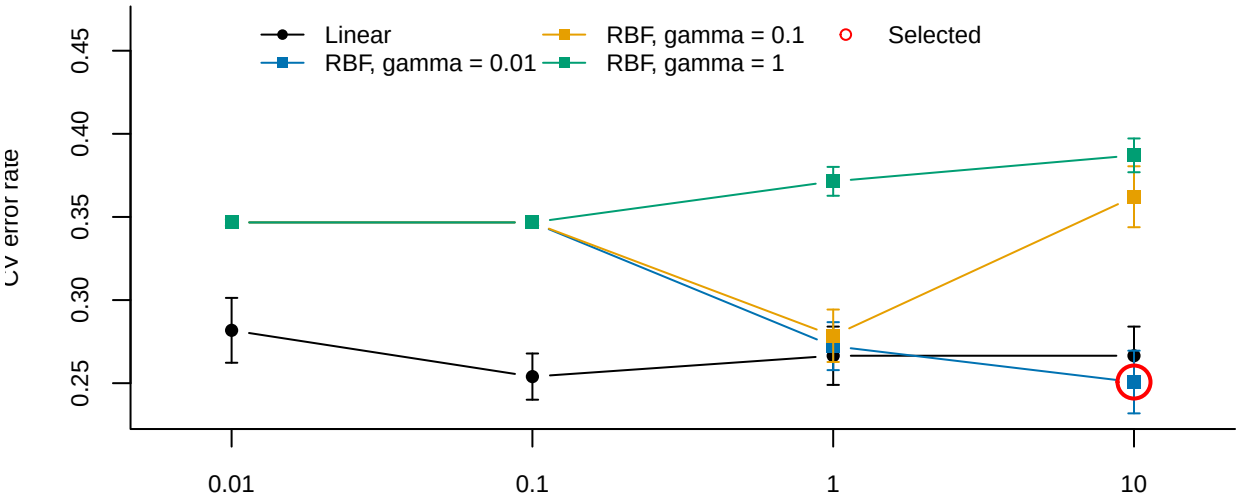

Table: CV misclassification rate (SE) for all candidates; selected model in bold.

C	Linear	RBF gamma=0.01	RBF gamma=0.1	RBF gamma=1
0.01	0.282 (0.020)	0.347 (0.002)	0.347 (0.002)	0.347 (0.002)
0.1	0.254 (0.014)	0.347 (0.002)	0.347 (0.002)	0.347 (0.002)
1	0.267 (0.018)	0.272 (0.014)	0.279 (0.016)	0.371 (0.009)
10	0.267 (0.018)	**0.251 (0.019)**	0.362 (0.018)	0.387 (0.010)

Results

The selected model is an RBF SVM with C = 10, gamma = 0.01 (CV error 0.251, SE 0.019). Its test error is 0.302 (42/139); sensitivity is 0.438 and specificity 0.835.

	Pred. 0	Pred. 1
Obs. 0	76	15
Obs. 1	27	21

Findings and limitations

− The best linear model (C = 0.1) had CV error 0.254 and the best RBF model (C = 10, gamma = 0.01) had 0.251; the gap of 0.003 is smaller than one SE (0.019).
− 4 of 16 candidates fall within one SE of the best, so CV alone does not clearly separate them; the lowest mean error was selected.
− Cost of false negatives: 27 of 48 test patients with CHD were missed (56.2%); in screening this is the costly error, and plain error rate does not weight it.
− The test set is small (139 cases, 48 with CHD): the SE is about 0.039 for the error and 0.072 for sensitivity.

Appendix: complete contents of SKILL.md

```
---
name: svm-model-fitting
description: Fit, tune, and evaluate Support Vector Machine (SVM) classifiers with rigorous
    cross-validation, explicit hyperparameter grids, and clinical diagnostic metrics.
---
# SVM Model Fitting & Evaluation Skill

## Workflow
1. **Suitability**: Verify that sample size and dimensionality are adequate. Standardize
    continuous predictors ($\mu=0, \sigma=1$) and encode categorical variables explicitly.
2. **Kernels**: Choose candidate kernels (Linear, RBF) and explain their theoretical
    justification based on decision boundary complexity.
3. **Tuning & Grid**: Define an explicit hyperparameter grid (e.g., $C \in \{0.01, 0.1, 1,
    10\}$; $\gamma \in \{0.01, 0.1, 1\}$). Use stratified $K$-fold cross-validation with a
    fixed random seed.
4. **Evaluation**: Select the final model via cross-validation error. Evaluate on a held-out
    test split, reporting test error, sensitivity, specificity, and a full confusion matrix
    to account for missed cases.
5. **Reporting**: Require a compact table listing every evaluated candidate model and a
    tuning visualization.
6. **Interpretation**: Report optimal parameters, numerical results, and the practical cost
    of false negatives.
7. **Page Budget**: Restrict the entire analysis output to a single printed page at 11-point
    font.
```